S5 MATHEMATICS TEST

DURATION: 1 1/2 HRS

INSTRUCTIONS

Attempt all items in this paper

ITEM 1

In a certain mathematical exploration, two students, Alex and Bella are studying the properties of two positive real numbers **a** and **b**. They discover that the relationship between those numbers satisfies the equation:

$$a^2 + b^2 = 23ab$$

Unfortunately they curious to explore further, they aim to prove a logarithmic relationship between **a** and **b**, specifically they want to validate their hypothesis that:

$$\log a + \log b = 2\log\left(\frac{a+b}{5}\right)$$

TASK:

As a mathematical student help Alex and Bella to prove their hypothesis.

ITEM 2

Mr. Omondi is organizing for a party in his ancestral home and would like to hire some vehicles to help in transporting the guests.

The number of vehicles to be hired is a product of the two values that satisfy the equation $\log_2 x - \log_x 8 = 2$ and each vehicle will be hired at **UGX 450,000**. In order to make his plan a success, he is to withdraw some money from centenary bank but he has forgotten the PIN. When he checked his diary, he formed a hint for the PIN and this was;

The first two digits satisfy the equation $9^x - 12(3^x) + 27 = 0$ and hand to be written in ascending order.

The next two digits are **a** and **b** which are obtained from expressing $\frac{1+\sqrt{3}}{2-\sqrt{3}}$ in the form $a+b\sqrt{3}$ such that in the PIN **a** takes position three and **b** takes the last position.

When he consulted a certain supplier, he was told that 4 sachets of salt, 3 Bars of soap and 2kg of sugar altogether cost **$8.2** and also 2 sachets of salt, 4 Bars of soap and 3kg of sugar cost altogether cost **$10.35**.

Similarly, 1 sachet of salt, 3 Bars of soap and 8kg of sugar will cost **$14.8**. Under keen analysis, Mr. Omondi's function requires that he must purchase 100 sachets of salt, 60 Bars of soap and 120kg of sugar.

His major intentions are to know the number of vehicles to be hired hence their total cost, generate his centenary bank account PIN and determine how much he should have to purchase salt, soap and sugar for his function.

TASK: Help Mr. Omondi to;

a) Determine with clear justification how much money he will spend on hiring the vehicles.

b) Generate his centenary account **PIN**.

c) Determine how much money he will spend for the purchase of salt, sugar and soap required for his party.

ITEM 3

In certain pharmaceutical company a certain product is produced by using a process function which depends on thickness, t of the tablet produced in mm given $f(t) = 2t^2 - 12t + 18$.

At a certain instant $f(t) = 0$, tablets are produced with one thickness, **V**. However the company started facing problems with the machine producing tablets with smaller thickness.

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Unfortunately the company has no money to import a new and a better machine but after the analysis, it has been observed that tablets produced differ from the previous one by **2mm**.

Your father is a process Engineer in this company and he is stuck on what to do and has approached you.

TASK:

As a student of S.5 mathematics: Use mathematical concepts to come up with a new process function that can be used by your father to continue with tablet production.

Help him to know the thickness of the tablet for which the function is minimum.